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1 Affine Varieties and Nullstellensatz

Claim: Let A be a Noetherian integral domain and $S \leq A$ a multiplicatively closed subset containing 1 and not containing 0. Then $B = S^{-1}A = \{a/s : a \in A, s \in S\}$ is again Noetherian.

Proof: Let $I \trianglelefteq B$ and $J = B \cap A$. I claim that $I = S^{-1}J$. Indeed, $S^{-1}J \leq S^{-1}I \leq I$ and if $a/s \in I$, then $a = s(a/s) \in J$, so $a/s = (1/s) \cdot a \in S^{-1}J$. $J \trianglelefteq A$. Let $\{I_n\}_{n \in \mathbb{Z}^+}$ be an ascending sequence of ideals of A and $\{J_n\}$ the corresponding J 's. Since A is Noetherian, let the J_n 's stabilize at $N \in \mathbb{Z}^+$. Then the I_n 's stabilize at N since $I_n = S^{-1}J_n$.

Claim (Hilbert basis theorem): If A is Noetherian, so is $A[x]$.

Proof: Let $J \trianglelefteq A[x]$. We show that J is finitely generated. Let I_n denote the set of leading coefficients of polynomials in J of degree n , along with 0. I claim that $I_n \trianglelefteq A$. Suppose $a, b \in I_n$, both nonzero. Then there exist $f, g \in J$ of degree n with leading coefficients a, b . If $f + g$ has degree n , then $a + b \in I_n$. Else, $a + b = 0 \in I_n$. Similarly, if $c \in A$ is arbitrary, then $ca \in I_n$ (depending on whether $ca = 0$ or not), so $I_n \trianglelefteq A$.

I claim that the sequence $(I_n)_{n \in \mathbb{Z}_{\geq 0}}$ is ascending. If $a \in I_n$ is nonzero, there exists $f \in J$ with degree n and leading coefficient a . Thus, $xf \in J$ has degree $n + 1$ and leading coefficient a , ie $I_n \leq I_{n+1}$. Let the ideals stabilize at N . Since A is Noetherian, let $\{a_{nm}\}_{m \in [r_n]}$ finitely generate I_n for all $0 \leq n \leq N$. For each a_{nm} pick a corresponding element $g_{nm} \in J$.

I claim that $\{g_{nm}\}$ generates J . First, $g_{nm} \in J$ for all n, m . Suppose $f \in J$ has degree l more than N . Let the leading coefficient of f be $a \in I_N$. Write $a = \sum_{i \in [r_N]} b_i a_{Ni}$. Then $f - \sum_{i \in [r_N]} b_i x^{l-N} g_{Ni} \in J$ has degree less than l . Suppose $f \in J$ has degree $l \leq N$. Write $a = \sum_{i \in [r_l]} b_i a_{li}$. Then $f - \sum_{i \in [r_l]} b_i g_{li} \in J$ has degree less than l . The base case is elements of degree less than 0, which is the element 0. Thus, $\{g_{nm}\}$ is a finite generating set of J .

Claim:

- (i). $V(0) = \mathbb{A}_k^n, V(A) = \emptyset$.
- (ii). $I \leq J \rightarrow V(I) \supseteq V(J)$.
- (iii). $V(I_1 \cap I_2) = V(I_1) \cup V(I_2)$.
- (iv). $V(\sum_{\lambda \in \Lambda} I_\lambda) = \bigcap_{\lambda \in \Lambda} V(I_\lambda)$.

Proof: (i). The first is good. For the second, take $x, x-1 \in A$, which have no common vanishing set.

(ii). Suppose $p \in V(J)$, so $f(p) = 0$ for all $p \in J$. In particular, $f(p) = 0$ for all $p \in I$, so $p \in V(I)$.

(iii). $V(I_1 \cap I_2) \geq V(I_1) \cup V(I_2)$ by part (ii). Suppose $p \notin V(I_1) \cup V(I_2)$, so there exist $f \in I_1, g \in I_2$ such that $f(p) \neq 0, g(p) \neq 0$. Then $fg \in I_1 \cap I_2$ such that $f(p)g(p) \neq 0$, so $p \notin V(I_1 \cap I_2)$.

(iv). $V(\sum_{\lambda \in \Lambda} I_\lambda) \leq \bigcap_{\lambda \in \Lambda} V(I_\lambda)$ by part (ii). Suppose $p \in \bigcap_{\lambda \in \Lambda} V(I_\lambda)$. For any λ , for all $f \in I_\lambda$, we have $f(p) = 0$. Suppose $g = \sum_{i \in [n]} g_i$ has $g_i \in I_{\lambda_i}$. Then $g(p) = \sum_{i \in [n]} g_i(p) = 0$, so $p \in V(\sum_{\lambda \in \Lambda} I_\lambda)$.

Claim (reverse correspondence):

(a). $X \leq Y \rightarrow I(X) \geq I(Y)$.

(b). For any subset $X \leq \mathbb{A}_k^n$, we have $X \leq V(I(X))$, equality iff X is algebraic.

(c). For $J \leq A$, $J \leq I(V(J))$.

Proof: (a). Let $f \in I(Y)$, so f vanishes on Y . In particular, f vanishes on X , so $f \in I(X)$.

(c). Each $f \in J$ vanishes on $V(J)$, so $J \leq I(V(J))$.

(b). Let $f \in I(X)$. Then f vanishes on X by definition, so $V(I(X)) \geq X$. Suppose X is algebraic, so $X = V(J)$ for some $J \leq A$. We have $J \leq I(V(J)) = I(X)$, so $X = V(J) \geq V(I(X))$. Since $V(I(X)) \geq X$ always, we have $X = V(I(X))$. Conversely, if $X = V(I(X))$, then X is algebraic, using the ideal $I(X)$.

Claim: Let $X \leq \mathbb{A}_k^n$ be algebraic and $I(X)$ the corresponding ideal. Then (a) X is irreducible iff $I(X)$ is a prime ideal. (b). Any algebraic set has a unique expression $X = \bigcup_{j \in [r]} X_j$ with X_i irreducible and $X_i < X_j$ for $i \neq j$.

Proof: (a). Let X be irreducible and suppose $f, g \in A$ such that $fg \in I(X)$. $V(f) \cup V(g) = V(fg) \geq V(I(X)) = X$, so either $V(f) \cap X = X$, ie $V(f) \geq X$ or $V(g) \geq X$, so $(f) \leq I(V(f)) \leq I(X)$, ie $f \in I(X)$. The other case is the same. Now suppose $I(X)$ is prime and $X = X_1 \cup X_2$ for algebraic X_1, X_2 . Then $I(X) = I(X_1) \cap I(X_2)$. I claim that $I(X) = I(X_1)$ or $I(X) = I(X_2)$. Else, pick $f \in I(X_1) - I(X)$ and $g \in I(X_2) - I(X)$. Then $fg \in I(X_1) \cap I(X_2) = I(X)$, absurd. WLOG $I(X) = I(X_1)$. Then $X = V(I(X)) = V(I(X_1)) = X_1$.

(b). The set of algebraic sets has the dcc, since we can apply I to get an ascending sequence of ideals in A (which is Noetherian), and then apply V . Let Σ denote the set of algebraic sets with no decomposition into irreducible components. If $\Sigma = \emptyset$, we're done. Else, $\Sigma \neq \emptyset$ has a minimal element, say X . If X is not irreducible, so there exist algebraic $X_1, X_2 < X$ such that $X = X_1 \cup X_2$. By minimality of X , X_1, X_2 can be written as a union of irreducibles. Thus, X is a union of irreducibles, showing existence.

If $X = \bigcup_{i \in [r]} X_i$ is a union of irreducibles and $X_i \leq X_j$ for some $i \neq j$, then we can remove X_i from the union. We have $I(X) = \bigcap_{i \in [r]} I(X_i)$, which in particular is a primary decomposition

of $I(X)$. Suppose FTSOC $I(X_1) \geq \bigcap_{i>1} I(X_i)$, so $I(X_1) \geq I(X_i)$ for some $i > 1$ (using that each $I(X_i)$ is prime), which gives $X_1 \leq X_i$, absurd. In general, $I(X_i) \not\geq \bigcap_{j \neq i} I(X_j)$, so the above primary decomposition of $I(X)$ is reduced. The primes of the reduced primary decomposition are unique (and they equal the $I(X_i)$), and the X_i are recoverable by $X_i = V(I(X_i))$, so the decomposition of X into irreducibles is unique.

Claim (Nullstellensatz): Let k be an algebraically closed field. Then:

- (a). Every maximal ideal of $A = k[X_1, X_2, \dots, X_n]$ is of the form $m_P = (X_1 - a_1, X_2 - a_2, \dots, X_n - a_n)$ for some $P \in \mathbb{A}_k^n$.
- (b). Let $J \subseteq A$, $J \neq A$. Then $V(J) \neq \emptyset$.
- (c). For any $J \subseteq A$, $I(V(J)) = r(J)$.

Proof: (a). (show later).

(b). Let $J \subseteq m$ for maximal m . Then $V(J) \supseteq V(m) = \{P\}$, where $m = (X_1 - a_1, \dots, X_n - a_n)$ and $P = (a_1, \dots, a_n) \in \mathbb{A}_k^n$.

(c). $I(V(r(J))) \supseteq r(J)$, and $V(r(J)) = V(J)$. Suppose $f \in I(V(J))$, so f vanishes on $V(J)$. Note that $f \notin k^\times$, so $(f) \neq A$.